

Web Application Servers

Web application servers are simple to define at a high level. They carry out the main business logic that processes a user's request and returns HTML to the browser. They often interface with a variety of backend infrastructure, including databases, caching layers, job queues, search services, other microservices, data/logging queues, and more, in order to perform their duties.

As previously stated, in order to process user requests, you normally have at least two, and frequently many more, load balancers plugged in. You should be aware that app server implementations necessitate the selection of a programming language (Node.js, Ruby, PHP, Scala, Java, C#.NET, and so on) as well as a web MVC framework for that language (Express for Node.js, Ruby on Rails, Play for Scala, Laravel for PHP, etc.).

Database Servers

To store data, every modern online application uses one or more databases. Databases allow you to define your data structures, input new data, retrieve existing data, update or delete existing data, execute cross-data computations, and more.

The web app servers, as well as the job servers, will most likely communicate directly with one another. Each backend service may also have its own database that is separate from the rest of the application.

While I'm trying to avoid going into too much detail about specific technologies for each architecture component, I'd be remiss if I didn't include the next level of detail for databases: SQL and NoSQL.

SQL stands for "Structured Query Language," and it was created in the 1970s to provide a universally accessible technique for accessing relational data sets. Data is stored in SQL databases in tables that are linked together by common IDs, which are usually integers.

Let's take a look at a simple example of preserving customer information. You might have two tables, customers and orders, that are linked by the user's id. A simplified version can be seen in the image below. Because the customer_id column in users table is a "foreign key" to the orders table, the tables are linked.

If you're not familiar with SQL, you can go through one of the many free courses from any of the MOOC (Massive Open Online Courses) providers. Because it's so common in web development, you'll want to master the basics at the very least in order to effectively construct an application.